THE UNIVERSITY OF MELBOURNE

FNCE 30007 Derivative Securities

Lecture 5 – Binomial Option Pricing Model I

Dr Xiaoting Wei

Review of Previous Lecture

Last week we

- Introduced options and key definitions
- Looked at payoffs and profits
- Discussed variables that affect their value
- Derived limits on option prices

What we didn't do:

- Derive a price for the option
- We did this for forwards and futures via no-arb arguments

Lecture Overview

In today's lecture we will discuss the Binomial Option Pricing Model. We will focus on:

- How we use a binomial tree approach to value European options

Introduction

Binomial Option Pricing Model

- Binomial option pricing is a simple but powerful technique that can be used to solve many complex option-pricing problems. It has proved to be one of the most flexible, intuitive and popular approaches to option pricing.
- In contrast to the Black-Scholes and other complex option pricing models that require solutions to stochastic differential equations, the binomial option pricing model is mathematically simple.
- It is based on the simplification that over a single period (of possibly very short duration), the underlying asset can only move from its current price to two possible levels.
- A very popular technique for pricing a stock option involves constructing a *Binomial Tree*.

A One-period Binomial Model

Assumption 1: Arbitrage opportunities do not exist

- We set up a portfolio in such a way that there is no uncertainty about the future value of the portfolio, so the return on the portfolio must be the risk-free rate

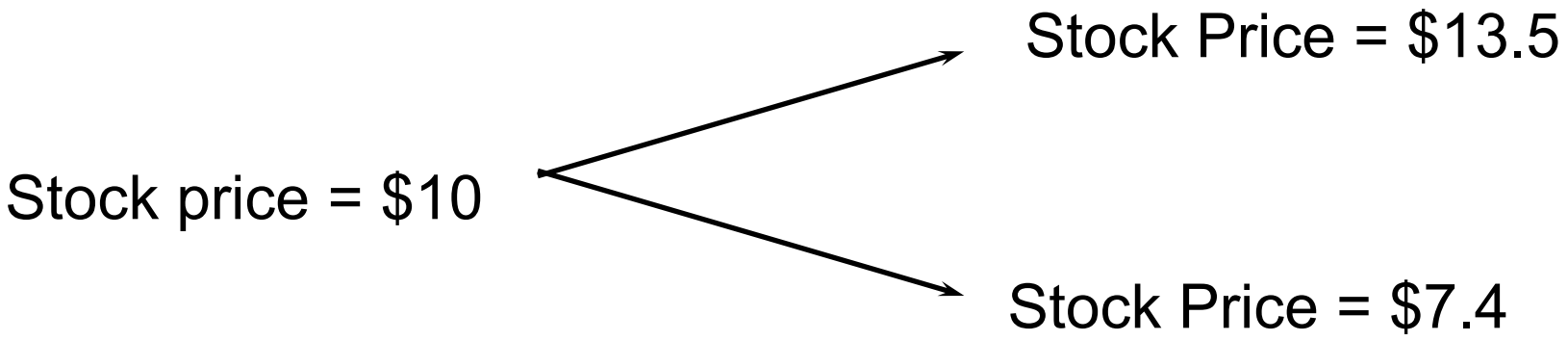
Assumption 2:

- The current stock price is S. In one period's time, the stock price will take on one of two values. It starts at S, then it either:
  - moves up, or
  - moves down
- This is what is meant by *Binomial* (only two events can occur).

A One-period Binomial Model

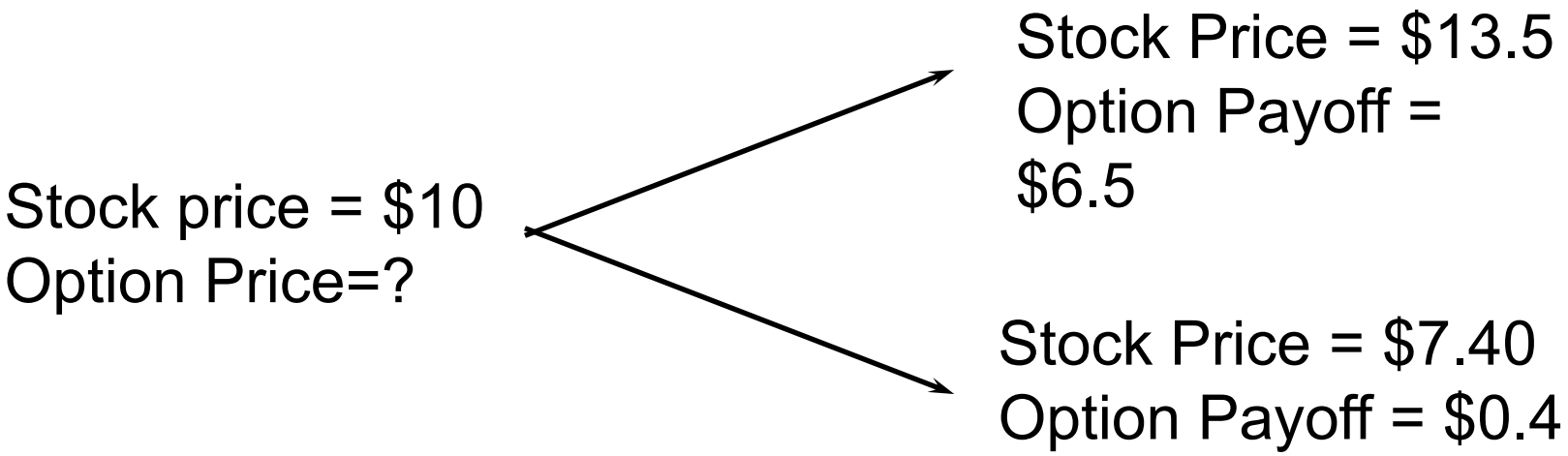
A stock price is currently $10; and,

In one-year it will be either $13.5 or $7.40.



A One-period Binomial Model

How would we value a one-year call option on the stock that has a strike price of $7? Assume a risk-free rate of 5% p.a.



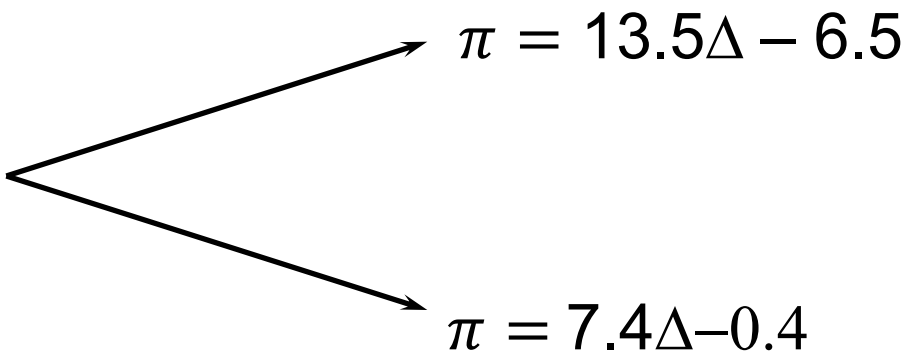
A One-period Binomial Model

- You might be considering estimating the probabilities of upward and downward movements, and then calculating the option price using the expected option payoff based on those probabilities
- Based on what we have learnt in Finance so far, that is a reasonable and correct approach (see slide 17). However, we want to develop a simpler and cleaner method that requires less work while still yielding the same answer.

A One-period Binomial Model

We aim to construct a riskless portfolio that has the same value whether the stock price is \$7.4 or \$13.5.

Consider the Portfolio: long Δ shares
short 1 call option



Portfolio is riskless when $13.5\Delta - 6.5 = 7.4\Delta - 0.4$ or $\Delta = 1$.

A One-period Binomial Model

The riskless portfolio is therefore:

- Long 1 share; and,
- Short 1 call option.

The value of the portfolio in one year is:

$$\pi = 13.5 \text{ x } 1 - 6.5 = 7 \text{ or } \pi = 7.4 \text{ x } 1 - 0.4 = 7$$

A One-period Binomial Model

How do we determine the correct position in the stock and the call option?

- Why not take a long position in both Δ shares and a call option?
- When constructing a riskless portfolio, we want the stock position to move in the opposite direction of the call position. Specifically, when the stock price rises, the value of a long stock position increases, while the value of a short call position decreases.
- By combining these positions, we can create a riskless portfolio whose value remains the same regardless of whether the stock price goes up or down.

A One-period Binomial Model

The portfolio that is:

long 1 share
short 1 option

} is worth $7e^{-0.05*1}$ =$6.6586 today

The value of the shares is:

$10 (= 1 x $10)

The value of the option is therefore:

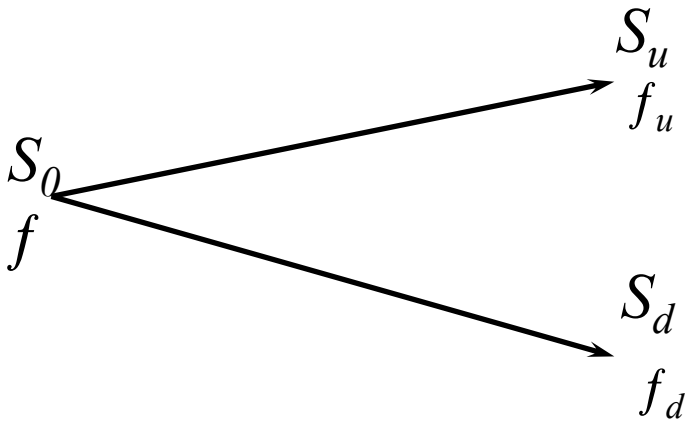
$10 – f = $6.6586

f = $3.34

A One-period Binomial Model

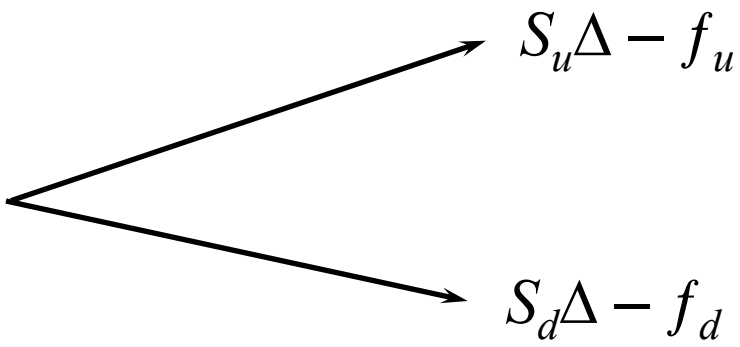
A generalisation of the previous example is provided below:

- A derivative lasts for time *T* and is dependent on a stock.
  - Where: Stock Price (S_0), Current Option price on Stock (f), Option duration on Stock (T), Option price increasing to new level (S_u), Option price decreasing to new level (S_d), Payoff from option when the stock price moves up (f_u) and payoff from option when stock price moves down (f_d).



A One-period Binomial Model

Consider the portfolio that is long position in Δ shares and short position in 1 option.



The portfolio is riskless when $S_u\Delta - f_u = S_d\Delta - f_d$ or:

$$\Delta = \frac{f_u - f_d}{S_u - S_d}$$

A One-period Binomial Model

Value of the portfolio at time T is:

$$\pi = \mathrm{S_u}\Delta - f_\mathrm{u}$$

Value of the portfolio today is:

$$\pi = (S_u\Delta - f_u)e^{-rT}$$

Another expression for the portfolio value today is:

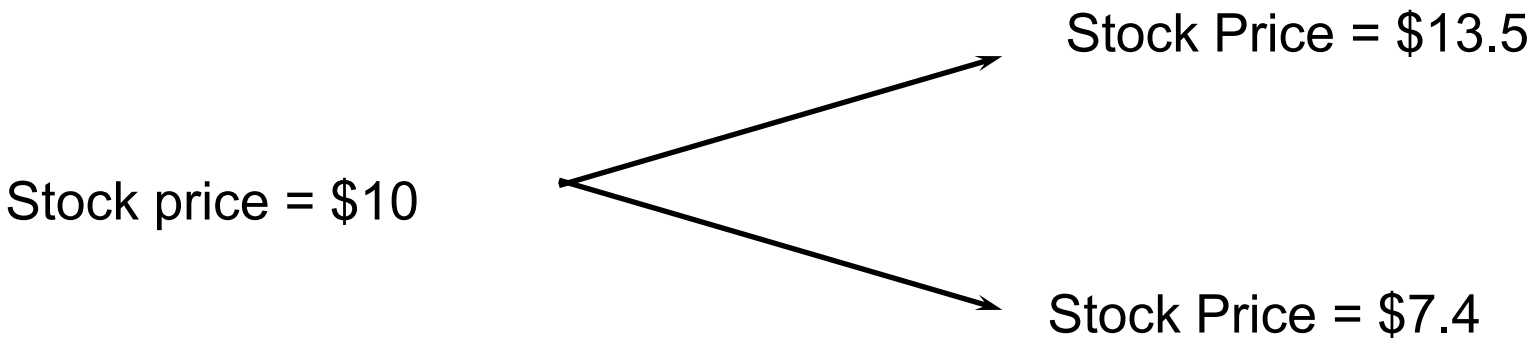
$$S\Delta - f$$

Hence:

$$f = S\Delta - (S_u\Delta - f_u)e^{-rT}$$

A One-period Binomial Model

How do we value a put option? Assume K=$13.5 and a risk-free rate of 5% p.a.



Traditional approach

Recall: fundamental valuation principle states that an asset's value equals its expected future cashflows discounted to present value at an appropriate discount rate.

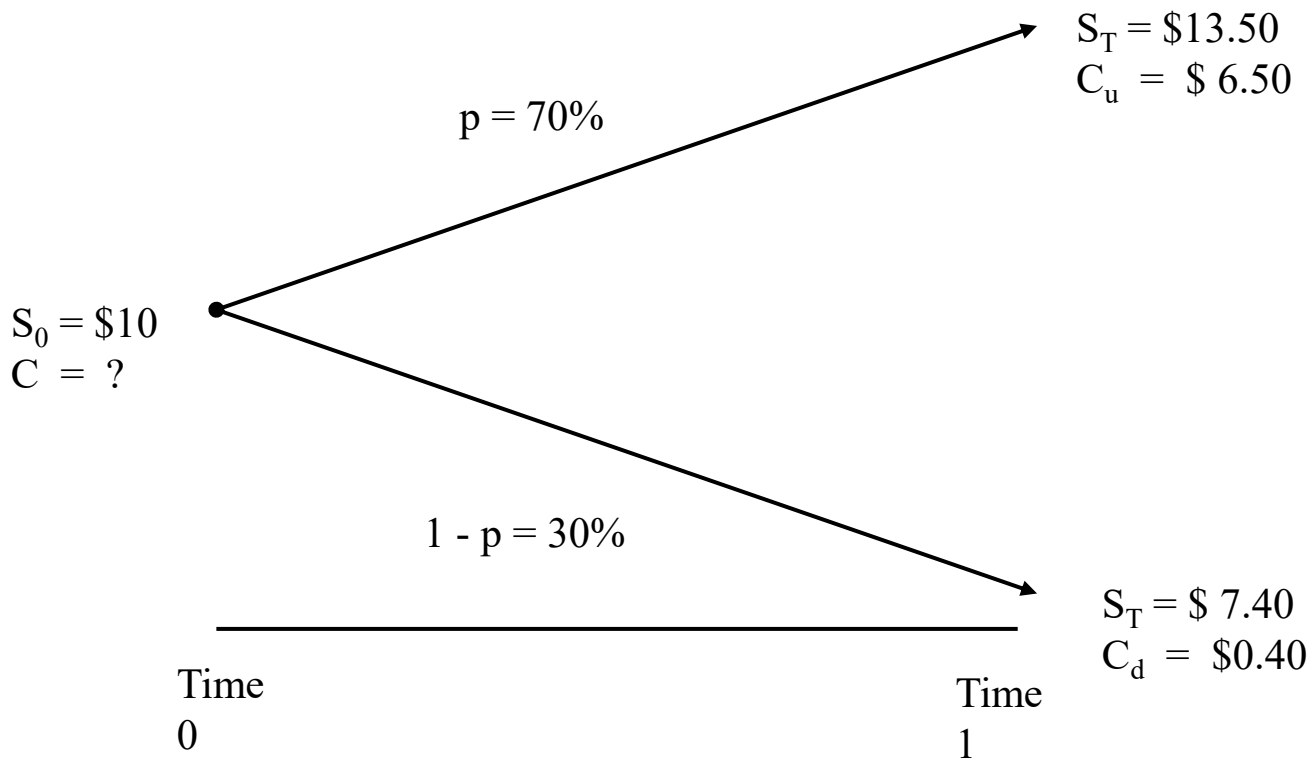
In $FNCE10002$ Principles of Finance, you valued many assets as follows:

- Draw a timeline and schedule the expected cashflows from the asset,
- Consider how risky/uncertain the cashflows are and choose a discount rate (a required rate of return) that reflects the riskiness,
- Discount all cashflows to present value and add them up to get a value.

This approach is used to value stocks, bonds, projects,and companies.

Traditional approach

Let's try to use this familiar approach to value a call option that had a $7 strike and one year to expiry.



Traditional approach

$$\text{ExpectedOptionPayoff} = (70\% \times \$6.50) + (30\% \times \$0.40) = \$4.67$$

$$E(S_1) = 0.7 \times 13.5 + 0.3 \times 7.4 = 11.67$$

$$E(r_S) = \frac{11.67 - 10}{10} = 0.167$$

What is the appropriate discount rate to reflect the riskiness of this call option?

Cox and Rubinstein (1985) show that the appropriate rate of return for a call option is:

$$\Omega = \frac{S}{C}\Delta$$

$$E(r_C) = r_f + \Omega\left(E(r_S) - r_f\right)$$

$$E(r_C) = 0.05 + \frac{10}{C}(0.167 - 0.05)$$

Traditional approach

If we use the traditional valuation approach, we take the expected option payoff and discount it to PV:

$$C = \frac{\text{Expected Option Payoff}}{1 + \text{E}(r_c)} = \frac{(70\% \times \$6.50) + (30\% \times \$0.40)}{1 + \left[0.05 + \frac{10}{C}(0.167 - 0.05)\right]}$$

This is one equation with one unknown (C), so a bit of algebra reveals the price of the option is $3.34 (same answer as in previous slides).

Therefore, the traditional approach does work for options. Why don't we use it?

Traditional approach

$$E(r_C) = r_f + \Omega(E(r_S) - r_f)$$

$$\Omega = \frac{S}{C}\Delta$$

The problem with the traditional approach is that the discount rate for the option depends on the current share price (S).

And since share price is constantly changing, this means that the appropriate discount rate for an option is also constantly changing.

If we examined a two-step Binomial tree, or three-step tree, we would find that the option discount rate is different on every branch of the tree!

When we valued stocks, or bonds or projects or companies, we never had the case where the required rate of return was changing through time!

Traditional approach

In previous slides, we valued a call option at $3.34 using a one-period Binomial tree. The most striking features of the no-arbitrage (delta-hedging) approach were that:

- We *did not use* the real-world probabilities of up/down movements in stock price (70%/30%).
- We *did not use* the expected return on the stock E(r_S), or the appropriate discount rate for the option E(r_C).

This is surprising! Given that the call option gives a bigger payoff ($6.50) if share price goes up and a smaller payoff ($0.40) if share price goes down, you'd think that the probability of up/down movements is crucial to valuing the option.

It turns out that this just isn't the case. And we can use this to our advantage ☺

Traditional approach

In fact, the probabilities of up/down movements are so irrelevant, that we can get the correct option value ($3.34) no matter what probabilities we use.

Let's use the probabilities of 60/40?

- The expected option payoff changes $(60\% \times \$6.50) + (40\% \times \$0.40) = \$4.06$
- The expected stock return changes $E(r_s) = (60\% \times 0.35) + (40\% \times -0.26) = 0.106$
- The option discount rate changes $E(r_C) = 0.05 + \frac{10}{C}(0.106 - 0.05)$

However, discounting the expected option payoff to present value still gives $3.34:

$$C = \frac{(60\% \times \$6.50) + (40\% \times \$0.40)}{1 + \left[0.05 + \frac{10}{C}(0.106 - 0.05)\right]} = \$3.34$$

Risk-Neutral Valuation

Asset pricing and valuation is difficult because investors are risk averse. They will want to be compensated by a higher expected return for holding a higher level of risk.

If, however, investors were risk-neutral, they would not demand extra compensation for bearing risk. The only discount rate we need to use would then be the risk-free rate.

We must find the probabilities that would apply if investors are risk-neutral.

The expected stock price in a risk-neutral world would be $S_0 e^{rT}$, therefore we have

$$pS_0 u + (1-p)S_0 d = S_0 e^{rT}$$

$$p = \frac{e^{rT} - d}{u - d}$$

Risk-Neutral Valuation

$p = \frac{e^{rT} - d}{u - d}$ where the u and d are the up/down movements proportional to the starting stock price:

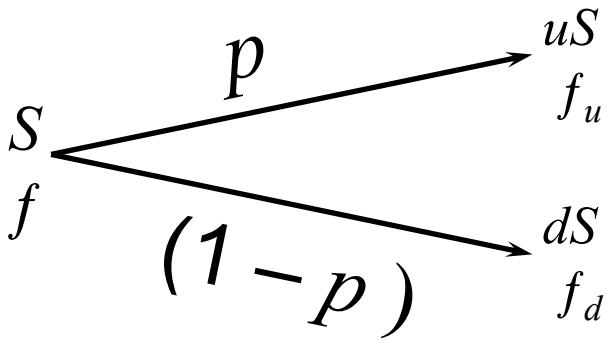
- *u* is the proportional increase (*u* > 1), e.g. u = 1.2 for a 20% increase in S
- *d* is the proportional decrease (*d* < 1), e.g. d = 0.7 for a 30% decrease in S

Risk-Neutral Valuation

The variables p and $(1-p)$ can be interpreted as the risk-neutral probabilities of up and down movements.

Stock price starts at *S*, then one period later, it has either (i) moved up to *uS*, or (ii) moved down to *dS*.

The value of an option is its expected payoff in a risk-neutral world discounted at the risk-free rate.



Option Value: $f = e^{-rT}[pf_u + (1-p)f_d]$

Risk-Neutral Valuation

$$p = \frac{e^{rT} - d}{u - d} = \frac{e^{0.05\times 1} - 0.74}{1.35 - 0.74} = 0.5103$$

$$1 - p = 1 - 0.5103 = 0.4897$$

$$\begin{aligned} f &= e^{-rT}[p * f_u + (1-p)f_d] \\ &= e^{-0.05\times 1}[(51.03\% \times \$6.50) + (48.97\% \times \$0.40)] \\ &= \$3.34 \end{aligned}$$

This is the same answer as the one calculated using the no-arbitrage method and the traditional method.

Risk-Neutral Valuation

So far, we have assumed values for *u* and *d*. In practice, *u* and *d* are determined from the stock price volatility.

Let Δt denote the length (in years) of one time step in the Binomial model.

Let σ denote the standard deviation of returns on the underlying stock.

Derive *u* and *d* as:

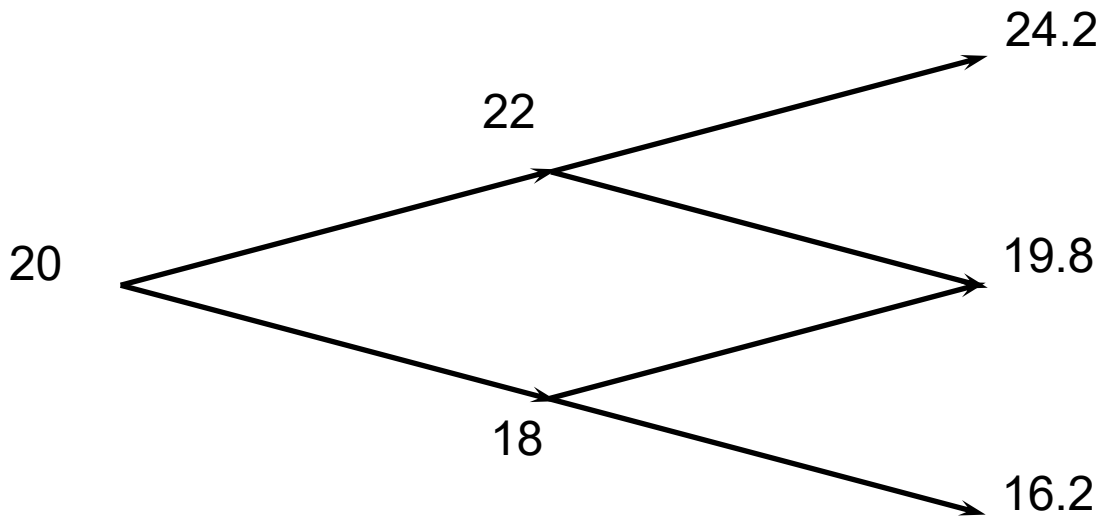
$$u = e^{\sigma\sqrt{\Delta t}}$$

$$d = \frac{1}{u}$$

Two-Step Binomial Tree Example

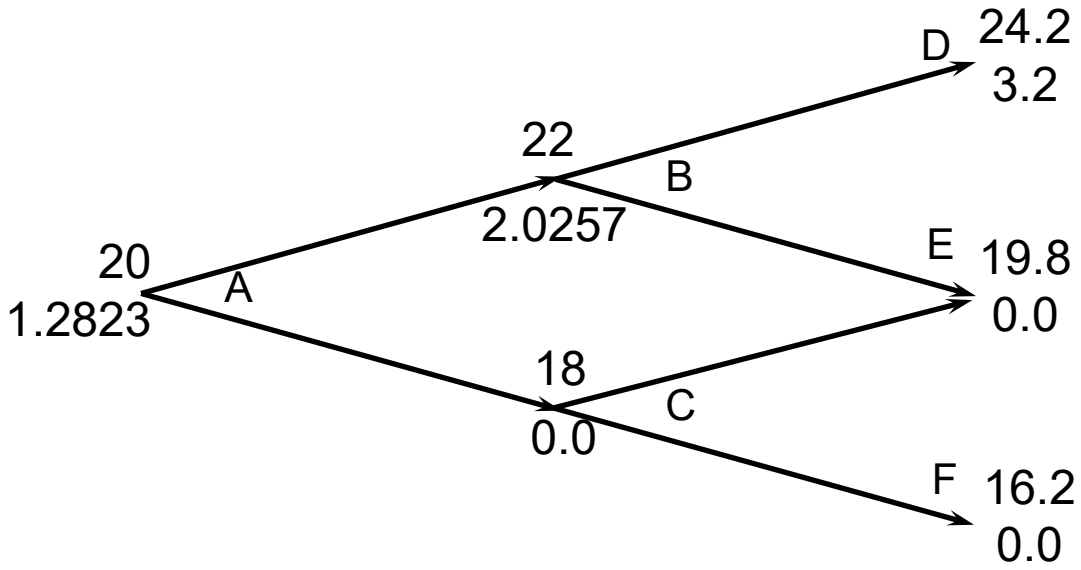
A call option has S_0 of \$20, K of \$21, r of 12% and σ of 19%. Calculate the price of the option with 6 months to expiration using a two-step binomial model.

$$u = e^{0.19*\sqrt{0.25}} = 1.1 \qquad d = \frac{1}{1.1} = 0.9$$



$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.12*3/12} - 0.9}{1.1 - 0.9} = 0.6523$$

Two-Step Binomial Tree Example



Value at node B:

$$= e^{-0.12 \times 0.25}(0.6523 \times 3.2 + 0.3477 \times 0) = 2.0257$$

Value at node A:

$$= e^{-0.12 \times 0.25}(0.6523 \times 2.0257 + 0.3477 \times 0) = 1.2823$$

Change of Delta over Time Steps

At the first time step (node A), $\Delta = \frac{f_u - f_d}{S_u - S_d} = \frac{2.0257 - 0}{22 - 18} = 0.5064$;

- If there is an upward movement over the first time step (Node B),

$$\Delta = \frac{3.2 - 0}{24.2 - 19.8} = 0.7273$$

- If there is a downward movement over the first time step (Node C),

$$\Delta = \frac{0 - 0}{19.8 - 16.2} = 0$$

This leads to portfolio rebalancing. We will cover this in more details in Week 10 – Greek Letters

Two-Step Binomial Tree Example: Valuing a Put Option

Consider a two-year European put with a strike price of \$52 on a stock whose current price is \$50.

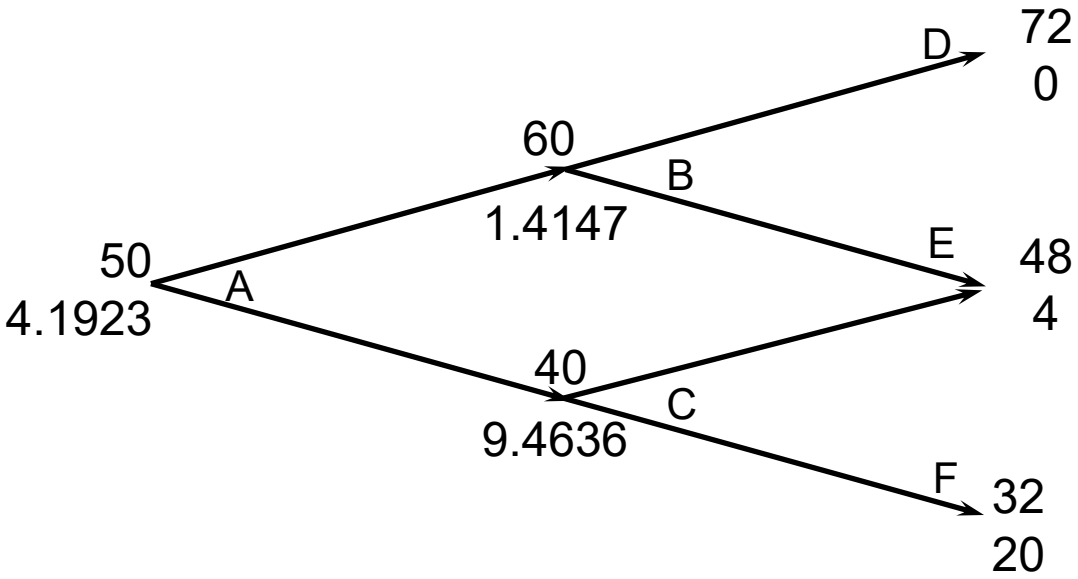
There are two time steps of one year each, and in each time step the stock price either moves up or down by 20%.

The risk-free rate of interest is 5% p.a.

In this case, u = 1.2, d = 0.8, $\Delta t = 1$ and r = 0.05.

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.05*1} - 0.8}{1.2 - 0.8} = 0.6282$$

Two-step Binomial Tree Example: Valuing a Put Option



The value of the put option is:

$$f = \frac{p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}}$$

$$f = e^{-2\times 0.05\times 1}(0.6282^2 \times 0 + 2 \times 0.6282 \times 0.3718 \times 4 + 0.3718^2 \times 20) = 4.1923$$

Next Week

- Binomial Option Pricing Model II
- Options on Futures